

Lab 4 Setup Guide

Building your Study Buddy environment with uv

1. Create a virtual environment with uv

Open a terminal in Cursor (Ctrl+`) and run:

```
uv venv .venv
```

2. Activate the virtual environment

```
# macOS/Linux
source .venv/bin/activate

# Windows
.venv\Scripts\activate
```

3. Install the required packages

```
uv pip install openai pypdf gradio python-dotenv ipykernel
```

- **openai** — the client library (includes AzureOpenAI)
- **pypdf** — reads the study notes out of notes/notes.pdf
- **gradio** — builds the chat interface
- **python-dotenv** — loads your .env file
- **ipykernel** — lets Cursor run the notebook using this venv

4. Select the notebook kernel

Open “Lab 4.ipynb”, click the kernel picker (top-right of the notebook), and choose the Python interpreter inside .venv.

5. Create your .env file

In the same folder as the notebook, create a file named .env:

```
AZURE_OPENAI_ENDPOINT=https://apim-azr-ue2-bgpt-prd-ucin.azure-api.net
AZURE_OPENAI_API_KEY=your-apim-subscription-key-here
```

Use the same APIM gateway endpoint and subscription key you used in earlier labs. Lab 4 always uses a single fixed deployment, gpt-5.4-ptu; you don't set that in .env, it's hardcoded in the notebook.

Never commit this file.

6. Add your study notes

In a notes folder next to the notebook, add a file named notes.pdf:

```
notes/notes.pdf
```

A sample notes.pdf covering the course itself is provided with this lab. You're free to swap in your own notes on any topic. The notebook only needs the file to exist at that path and contain extractable text, not a scanned image.

7. Run the notebook

Run the cells top to bottom (the Run button, or Shift+Enter). When you reach a `gr.ChatInterface(...).launch(inbrowser=True)` cell, a browser tab opens automatically, chat there, then use the cell's stop button (or Kernel → Interrupt) before moving to the next cell.

8. Troubleshooting

- **"Missing settings" error on Step 5** — your `.env` file is missing, misnamed, or in the wrong folder. It must be named exactly `.env` and sit next to the notebook
- **FileNotFoundError on notes/notes.pdf** — check the folder is named `notes` and the file is named `notes.pdf`, both lowercase
- **Gradio says the port is already in use** — an earlier `gr.ChatInterface(...).launch()` cell is still running, interrupt it, or restart the kernel, before launching a new one
- **ModuleNotFoundError for openai, pypdf, gradio, or dotenv** — re-run Step 3's install command and confirm the notebook's kernel is this project's `.venv`
- **An error partway through a multi-tool-call conversation** — if you're on Step 11 of the notebook (the `if-only` version of `chat()`) and it errors after asking it to log two things at once, that's expected. It's the lab demonstrating why Step 12 changes `if` to `while`, and it resolves by Step 13